Teacher answers and topic guidance

Accept scientifically equivalent answers in the learner's usual mode. Award each distinct point once. Keep answers hidden during independent attempts.

Starter

1. Active transport.
2. Water.
3. The nucleus.

Full and faded sequence

Interphase; prophase; metaphase; anaphase; telophase; cytokinesis. The faded step is anaphase: the two copies of each chromosome are pulled to opposite ends. DNA is copied before mitosis, in interphase.

Six-chromosome hinge

C: six in each daughter, with the same set as the original body cell. A, three: confuses this ordinary body-cell model with chromosome halving. B, twelve: treats copied material as the number retained by each daughter. D, none: leaves the copies in one cell instead of distributing them.

About three in four secure responses: move on. Otherwise re-model with two blue and two orange paper strips, labelled A and B, then use the parallel case. A starting set of eight gives eight in each daughter.

Science scope and misconceptions

Mitosis separates copied chromosomes into two nuclei. Cytokinesis splits the cell. The model concerns ordinary body-cell division. It does not teach the full molecular detail, asexual reproduction or chromosome halving. Do not describe mitosis alone as the whole final split.

Copy before separation: both new nuclei need a matching set. Uneven separation can leave one nucleus with an extra copy and another lacking it. New matching body cells support growth and replacement of damaged cells.

Arrival Supported

1. Two cells.
2. The one cell on the left.
3. The inner outline labelled nucleus.
4. One cell has become two; each new cell has a matching labelled set.

Arrival Independent

1. Two.
2. The nucleus.

3. Both cells have A and B: the sets match.

4. One cell became two, each with a matching chromosome set.

Arrival Aspire

1. Two separate cells.

2. The nucleus.

3. Each new cell has both types, A and B, so the sets match.

4. It shows one cell becoming two with matching sets; it does not show the full molecular detail or every real chromosome.

Exit Supported

1. Interphase, mitosis, cytokinesis.

2. Matching sets.

Exit Independent

1. Mitosis.

2. DNA was copied before mitosis and the matching copies separated into the two new nuclei.

Exit Aspire

1. The copies must exist before matching sets can be separated into two nuclei.

2. The DNA first has to be copied and the copies separated into matching sets; splitting alone does not distribute matching sets.

Made by Matt

Mark scheme Supported

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Order interphase, mitosis and cytokinesis. Show A and B in each new cell. [3]

Interphase, mitosis, cytokinesis; matching A and B sets in both new cells. Award one point for each correctly placed stage, with matching sets checked as the success criterion.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Finish: Each new cell matches because DNA was ___ and the copied chromosomes ___ into two nuclei. [3]

DNA copied; copied chromosomes separated; each new nucleus receives a matching set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. How do copy, separate and split work together when new cells are needed for repair? Show it or give one short explanation. [3]

DNA is copied; matching sets separate before the cell splits; the matching new body cells help replace damaged cells.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Independent

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Order interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Mark the four phases that belong to mitosis. [4]

Interphase first; prophase then metaphase; anaphase then telophase; cytokinesis last and the four middle phases identified as mitosis.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain why each new body cell has a matching chromosome set. [3]

DNA copied in interphase; matching copies separate during mitosis; each new nucleus receives a matching set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Use the model to explain how cell division helps replace damaged body cells. [3]

DNA copied; matching sets separated and cell split; new matching body cells can replace damaged cells.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Aspire

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Predict what would happen to the sets if copied chromosomes did not separate evenly. Give a reason. [3]

New nuclei could have unequal sets; one could gain a chromosome copy and another lack it; the sets would no longer match because separation was uneven.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain why swapping mitosis and DNA copying would break the model. [3]

DNA copying belongs before mitosis; without copies there is not a copied set to distribute to each new nucleus; both new cells would not receive the intended matching set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Fault finder: compare two faulty models. One skips DNA copying; the other puts both copies of one chromosome in one nucleus. Find each first error and propose a repair. [4]

Identify copying skipped in interphase; add the copying step before mitosis; identify both copies sent to one nucleus during separation; send one copy of each chromosome to each new nucleus before the final split.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Choice

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Choose the first, middle and last stage from interphase, mitosis and cytokinesis. Show both A and B in each new cell. [3]

Interphase first, mitosis middle, cytokinesis last, with matching labelled sets checked.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. A body-cell model starts with six chromosomes. Choose three, six or twelve for each daughter cell after ordinary mitosis and cell division. Give a reason. [3]

Six; DNA was copied before mitosis; matching copies were separated so each new cell keeps the original set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Choose more matching body cells or fewer chromosomes as the useful result for repair. Explain one link. [2]

More matching body cells; these can replace damaged cells.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt